Po Wei (Sander) Hsu

BBA, Information Management, National Pingtung University, 2019 - 2023

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BRIEF INTRODUCTION

Experienced AI and software engineer with 3+ years building intelligent systems. Proficient in CV, NLP, ML, and data engineering using Python, Golang, PyTorch, and LangChain. Equipped with a strong QA mindset and hands-on experience in test automation (Cypress, Playwright), ensuring reliability and quality across the development lifecycle. Skilled in end-to-end pipeline development, cloud deployment, and CI/CD. Collaborative and detail-oriented, with a focus on building robust, production-ready systems.

3+ years of experience in Software Development.
1+ years of experience in CV & LLMs.
1+ years of experience in ML & MLOps.
1+ years of experience in data engineering.
1+ years of experience in Software Quality Assurance.

SIDE PROJECTS & HOBBIES

- object-distance-estimator

A tool for computing the absolute distance of objects from the camera in video frames

- openhands-resolver-testrepo

A GitHub Action powered by an AI agent that quickly resolves issues by generating pull requests or fix suggestions.

- Recommendation System Experiment Notes

A recommendation system project where I implemented several classic recommendation algorithms.

- Product Technology Demo of 3D Volume Rendering Applied to E-Commerce Platform

A technology that makes product displays on e-commerce platforms more engaging by converting 2D images into 3D objects viewable directly on the web.

PERSONAL EXPERIENCES

AI Software Engineer, Ubiquiti , Taipei/Taiwan, Dec 2024 - Present

- **community-pipeline-Core & community-test-designer & community-api-services & grafana-chat-panel**

Python / FastMCP / LangGraph / RAG / Grafana / Airflow / Openai API / MongoDB / PostgreSQL / Prompt, Pipeline Engineering / Docker

Developed a fully automated monitoring and suggestion system based on UI-Community, achieving a 1000× efficiency boost. Also doubled the productivity of the Protect, Drive, Access, Connect, Switch and Play product teams by leveraging a custom-built TestDesigner Agent to address core PM and QA pain points.

- **Label Studio & dataset-faster-reviewer**

Golang / Gin / Next.js / Restful API / JWT / Swagger / Sqlite / PostgreSQL / Label Studio / Data Version Control (DVC) / Docker

Developed a full-stack app hooked with Label Studio to make dataset checking easier, reduced data review time by 30%. This setup keeps detailed metadata—like timestamps, statuses, and who did what for both checking and marking tasks—and works on a server to get rid of extra data moving around, making things 50% more efficient.

- **Thumbnail CV Tools**

C / C++ / Python / PyTorch / Cuda / TensorRT / DALI / Ultralytics

Developed an AI pipeline dedicated to processing thumbnails, boosting throughput from 1,000 to 10,000 images per second by leveraging NVIDIA DALI for GPU–CPU parallel processing. Integrated TensorRT and CUDA to accelerate inference, and applied denoising techniques that improved accuracy by 10%. The system fully replaced manual review, reducing human effort by 100%, and enhanced model score quantification efficiency.

- **Other Projects: Vision Weather & PR-Issue LLM Agent & AI-QA Automation Tools & Edge-AI Test Planning**

AI Software Engineering Intern, Pegatron Corporation , Taipei/Taiwan, May 2024 - Nov 2024

Python / Next.js / FastAPI / LangChain / Llama / RAG / Milvus / MongoDB / MariaDB / Chunk, Prompt, Pipeline Engineering

Engineered and streamlined internal LLM processes using Llama and LangChain, constructed scalable data pipelines with Milvus, MongoDB, MariaDB, and improved data pre-processing through semantic chunking, enhancing recall by 5%. Created interactive multi-modal chatbot response visualizations, participated in the design of a Mixture of Agents architecture that improved system performance by 20%, and worked cross-functionally to provide a frontend interface focused on user experience.

SQA Automation Engineering Intern, SWAG , Taipei/Taiwan, Sep 2024 - Nov 2024

Node.js / Cypress / Github Action / TDD, BDD, ATDD / Test Planning

Created and developed a customized automated testing framework from scratch utilizing Cypress and JavaScript, independently creating the test architecture, reusable components, and authoring composite UI and functional test suites for multi-layered verification of workflows, API integrations, and edge-case scenarios. Further, integrating these tests to CI/CD pipelines for seamless execution provided over 90% automation coverage while reducing manual testing effort by over 70%.

Research Assistant, National Pingtung University , Pingtung/Taiwan, Apr 2023 - Present

Golang / Network Protocol / Algorithm Development / Research Skills

Supervised by Professor Ching-Chih Chuang. Conduct research on TSN and the IIoT, focusing on deterministic communication and hybrid network optimization. Participated in projects: *"Deterministic Multicast Technology in Industrial Internet of Things"*, *"Optimization of Hybrid Network Architecture for the Industrial Internet of Things"*, and *"Study on Cross-Domain Technologies in TSN–CAN Hybrid Network Architectures"*. Contributed to improving reliability and efficiency in industrial communication systems through TSN-based designs and cross-domain integration.

Founder, 電到系列科技有限公司 / 電到創商有限公司 , Kaohsiung/Taiwan, Apr 2022 - Sep 2023

Leadership / Management / Decision-Making / Risk Tolerance / Creativity